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Scoping Review

# Adherence to Strength Training and Lower Rates of Sports Injury in Contact Sports

## A Systematic Review and Meta-analysis

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**Background:** Injury prevention is very important for athletes, and studies have been carried out to better inform effective means of injury prevention. Strength training has long been recognized as important for preventing sports injuries. But until now, there has been debate about the proper application of strength training to prevent injuries, especially for different types of injury.

**Purpose:** To compare the effects of single-component and multicomponent strength training on the prevention of different types of injuries and to provide reference for the future design of injury prevention programs (IPPs).

**Study Design:** Systematic review; Level of evidence, 1.

**Methods:** Five databases were searched in August 2024. The main inclusion criteria were randomized controlled trials on use of an IPP that included strength training with a primary outcome of sports injury rate. The initial search resulted in 3583 articles, which were filtered to 16 articles that met the inclusion criteria. Extracted data were subjected to meta-analysis using a random-effects model, and subanalysis was carried out to explore the efficacy of different component strength IPPs in mitigating injuries at different locations in the body.

**Results:** The pooled results based on total injuries showed that strength IPPs had a statistically significant reduction in injury relative risk (RR) of 0.70 (95% CI, 0.60-0.82). Subgroup analysis showed that single-component strength training significantly reduced groin injuries (RR, 0.69; 95% CI, 0.51-0.93) and hamstring injuries (RR, 0.37; 95% CI, 0.25-0.55). In addition, multicomponent strength training significantly reduced knee injuries (RR, 0.71; 95% CI, 0.51-0.98) and ankle injuries (RR, 0.68; 95% CI, 0.52-0.89).

**Conclusion:** This study demonstrates that different-component strength IPPs have differential preventive effects on sports injury because of different target sites. The reason may be that the anatomic structure and motor function of the target sites are different. We can determine the vulnerable locations according to the different characteristics of the sports and select different component strength IPPs according to the characteristics of the target sites.

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**Keywords:** strength training; sports injury; injury prevention; training modality; training load

When athletes sustain injuries and cease training for recovery, their original training plans are disrupted, hindering further advancements in athletic performance and significantly affecting expected competitive outcomes.[46] Consequently, the prevention of sports injuries has been a subject of extensive research, with existing evidence highlighting the critical role of training in mitigating such injuries. However, merely increasing the volume of training is insufficient to reduce the incidence of sports injuries, because there are many factors that contribute to the occurrence of sports injuries, including improper movement patterns, inappropriate loading, and excessive fatigue, among others.[2,39,51] As a result, only targeted training can reduce the risk of injury. Notably, recent research highlights the significant and essential role that strength training plays in injury prevention.[7]

There has been extensive research in strength training that focuses on biomolecular aspects, fall prevention in the elderly, general health gains, or specific musculoskeletal injuries.[15,20,30,38] Given the potential of strength training to enhance biomechanical efficiency, improve neuromuscular

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control, and mitigate limb asymmetry, as well as other pathways[6,22,59] that collectively contribute to the reduction of sports injuries, it is imperative to conduct research on the effect of strength training in minimizing the incidence of such injuries. Research in the domain of sports injury prevention indicates that strength training aimed at improving athletes' eccentric strength, core stability, and muscular balance can substantially reduce the risk of sports-related injuries.[18,28,44] While various modalities of strength training exist, studies have generally categorized them into 2 distinct types: single-component and multicomponent strength injury prevention programs (IPPs). Single-component strength IPPs include only 1 training component, and multicomponent strength IPPs combine strength with other training components such as agility, proprioception, and coordination, among others.[50] There has long been a distinction between single-component and multicomponent injury IPPs, and both have been studied for their effects on sports injury prevention.[14,36] Among these, we found that the effectiveness of multicomponent versus single-component IPPs has been a subject of considerable debate. According to some researchers, multicomponent IPPs reduce the rates of lower limb injuries more effectively.[8,14] As a good example, the FIFA 11+program has gained recognition for its integration of strength, plyometrics, speed, agility, and flexibility training into warm-up routines, resulting in a significant reduction in sports injuries during both matches and training sessions.[29,55] On the contrary, Lauersen et al[35] claimed that it is necessary to shift from multicomponent strength IPPs to single-component strength IPPs to prevent sports injuries, and a 10% increase in strength training volume reduced the risk of injury by >4 percentage points.

Different researchers hold differing opinions about the same research question; thus, we need to study what causes the controversy, starting with the mechanisms and principles. As Lauersen et al[35] point out, exercises can be claimed to be site-specific and injuries of interest to be sport-specific. McBain et al[40] have classified sports into collision, contact, and noncontact categories based on the inherent physicality of the sport, with high-risk sports typically carrying higher costs associated with sports injuries. Based on the inherent physicality of the sports, McBain et al have defined the concept of contact sports as sports with frequent contact but few collisions. Specific injury sites and types are related to the unique characteristics of each sport's movements and techniques,[15] such as knee and ankle injuries in basketball, shoulder injuries in baseball, and groin injuries in soccer, all 3 of which belong to the category of contact sport.[39]

This study aims to investigate the prevention of sports injuries in different parts of the body through strength training, to discuss site-specific exercises and sport-specific injury risks, and to summarize the effects and applications of different single-component and multicomponent strength IPPs.

## METHODS

### Search Strategy, Study Selection, and Data Extraction

The research protocol adhered to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and was registered on the PROSPERO platform, comprising a priori specification of analyses, study inclusion/exclusion criteria (Table 1), and complete search strategy (see Supplemental Material, available separately). Literature searches were conducted across multiple databases, including PubMed, Web of Science, SPORTDiscus (Ebsco), and Embase, spanning from inception to August 25, 2024, as shown in the study selection flowchart (Figure 1). The search contained 4 blocks of keywords related to strength training, prevention, injury, and randomized controlled trials (RCTs) combined with "AND."

The literature was initially screened by 2 researchers (J.W. and K.Z.) through a review of titles and abstracts based on predetermined inclusion criteria, with subsequent evaluation of the full text. Discrepancies between the researchers were resolved through discussion. Relevant data from the literature were extracted using a self-designed data extraction form, which included information such as author, year of publication, participant demographics (age, gender, specialty), sample size, details of intervention measures (duration, frequency), exposure duration, type and characteristics of injuries, injury probability, and quantity. The injury rate is equal to the number of injuries divided by the total number of subjects multiplied by 100%.

### Quantitative Analysis

Meta-analysis relative risk(RR) estimates, including different injury types, were calculated in R Version 4.2.3 (R Development Core Team). The outcome indicators of the study reported injury rates and sample sizes. We calculated the RR value and 95% CI of the experimental group and the control group by the total number of injuries in each group divided by the sample size, and combined them. The random-effects model was used for analysis, and subgroup analysis and sensitivity analysis were

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Ethical approval was not sought for the present study.

TABLE 1
Summary of Study Inclusion and Exclusion Criteria and Categories

| Item | Terms |
|---|---|
| Inclusion criteria | Randomized controlled trials; strength training interventions, adequate/sufficient volume, intensity, and character of intervention; amateur or professional athletes who regularly train ≥3 times per week, identify with a specific sport, and train for competition; outcome reports number or rate of injuries |
| Exclusion criteria | Nonrandomized controlled trials, intervention elements other than strength training, unclear interventions, incomplete outcome data, unknown injury type |
| Study design | Randomized controlled trials |
| Intervention type | Single component: injury prevention program includes only strength training component, designed to prevent the occurrence of sports injuries in specific body regions |
| | Multicomponent: injury prevention program consists of a variety of training components, including strength, sensitivity, proprioception, and plyometric exercises, designed to prevent the occurrence of common multiregion injuries |
| Injury location | Ankle, knee, shoulder, hamstring, groin |
| Injury type | Contusion-laceration-abrasion; fracture–bone stress; joint (nonbone)-ligament (eg, sprain, cartilage, patellofemoral syndrome); muscle-tendon; other |
| Contact sports Included | Handball, soccer, baseball, floorball, basketbal |

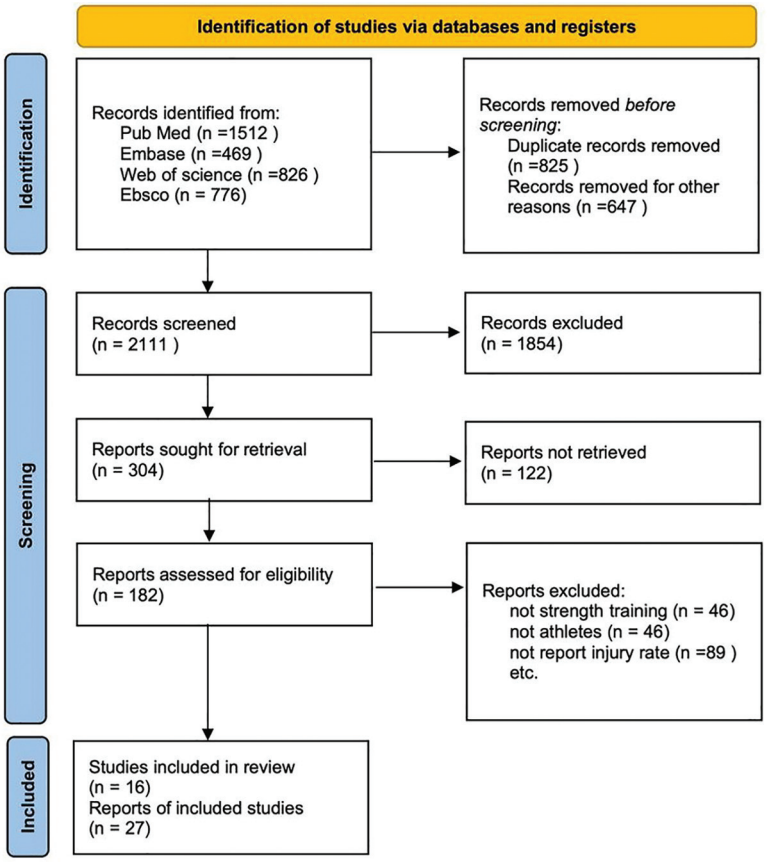


**Figure 1.** Study selection flowchart.

conducted according to the research objectives. Based on the research objectives, the subgroup analysis was conducted on the effects of strength training on reducing contact-sport injuries in different body parts, including injuries to the shoulder, knee, ankle, groin, and hamstring.

## Quality Assessment of Literature

The risk of bias in the included literature was evaluated according to the Cochrane risk-of-bias assessment tool, including 7 bias assessments toward literature quality.

The funnel plot method and the Egger test was used to test publication bias for studies.

## RESULTS

### Included Studies and Their Basic Characteristics

This study finally included 16 studies, with a total sample size of 7459 individuals, aged from 16 to 47 years old, with male patients accounting for >70%. To minimize the effect of insufficient training volume on the actual reduction of sports injury incidence, only individuals engaged in systematic amateur or professional training were included. Moreover, all included studies were randomized controlled trials, whose methodological quality was assessed by a quantitative quality assessment.

### Interventions and Compliance

Although all 16 studies designed interventions based on existing methodology and principles, there were a number of differences in the strength training programs. First, the modalities of these strength training programs included eccentric training (eg, Nordic hamstring exercises), concentric training (eg, shoulder external rotation muscle strength training, adductor strength training), isometric training (eg, isometric hip adduction training), and combined training (eg, ankle isometric and dynamic resistance strength training). Second, some strength training programs involved multiple components (eg, a combination of strength, balance, and plyometrics). These strength training programs were usually performed as a warm-up before training or competition. Finally, the intervention duration ranged from 10 to 46 weeks, with outcome indicators reporting overall or site-specific injury rates. The compliance reported in the studies was relatively high, with most exceeding 70%. Table 2 introduces basic information on the included studies. Table 3 introduces an overview of the intervention characteristics, including the interventions, compliance, exposure time, frequency, sets, and repetitions (Table 3).

### Meta-analysis and Subgroup Analysis Results

*Effect of Strength Training in Reducing Overall Sports Injury Rates.* Sixteen studies with 27 outcomes were included in the meta-analysis evaluating the effectiveness of strength training in reducing injury rates in contact sports (Figure 2), and the meta-analysis results showed low heterogeneity among studies ($I^2$ = 38%). Using a random-effects model, the combined RR value and 95% CI were 0.70 (0.60-0.82), indicating a 30% reduction in sports injury rates. No asymmetry was found in the funnel plot (Figure 3), and the Egger test confirmed symmetry ($t = -1.87$; SE = 0.4488; $P$ = .07), indicating no significant publication bias.

*Effect of Single-Component Strength Training on Reducing Groin and Hamstring Injuries.* Subgroup analysis showed that single-component strength training significantly reduced the incidence of groin and hamstring injuries; the groin injury rate decreased by 31% (RR, 0.69; 95% CI, 0.51-0.93) (Figure 4), and the hamstring injury rate decreased by 63% (RR, 0.37; 95% CI, 0.25-0.55) (Figure 5). The heterogeneity tests showed no heterogeneity in the injury rates of groin and hamstring (both $I^2$ = 0%).

*Effect of Multicomponent Strength Training on Reducing Injury Rates in Specific Body Parts.* Subgroup analysis showed that multicomponent strength training significantly reduced the incidence of ankle and knee injuries; the ankle injury rate decreased by 32% (RR, 0.68; 95% CI, 0.52-0.89) (Figure 6), and the knee injury rate decreased by 29% (RR, 0.71; 95% CI, 0.51-0.98) (Figure 7). The heterogeneity tests showed no heterogeneity in the ankle injury rates ($I^2$ = 0%) and low heterogeneity in the knee injury rates ($I^2$ = 47%).

*Effect of Strength Training on Reducing Shoulder Injuries.* The meta-analysis showed that 2 modalities of strength training did not reduce the incidence of shoulder injuries; all upper 95% CI values exceeded 1.00, indicating no apparent effect. The heterogeneity tests showed high heterogeneity in the shoulder injury rates ($I^2$ = 61%) (Figure 8).

### Quality Assessment of Included Studies

All 16 included studies were randomized controlled trials, with evidence quality at level 2 or higher. Most studies had no issues with random sequence generation, and some studies with large sample sizes were able to achieve good allocation concealment. However, in nonpharmacological trials, blinding of participants posed significant challenges, so some of the studies included in this research may not have completed true blinding, which could have influenced the results. Despite exclusion of studies with missing key outcome indicators because of differences in research questions and objectives, some indicators were not fully obtained, such as potential risk factors for injury at baseline. In addition, some studies used self-administered questionnaires or online surveys to assess injuries, which may have differed from injuries assessed by health care professionals in clinic. However, this does not significantly affect the results because the combined indicator in this study is the RR value. Several studies used questionnaire-based methods to record injuries[1,23,41,49]; these methods were limited to overuse injuries, and overuse injuries were uniformly investigated through questionnaire surveys; thus, results were not significantly affected. Specific literature quality assessments of the included studies are detailed in Figure 9.

## DISCUSSION

### Strength Training Reduces Injury Rates in Contact Sports Events

This study mentioned the frequent incidence of several site injuries in contact events, analyzed the influence of

TABLE 2
Basic Information of All Included Studies

| First Author, y | Team Sex[a], Age | Sports Event | Sample Size (Exp, Con) | Assessment Method | Injury Type |
|---|---|---|---|---|---|
| Andersson, 2017[4] | 23 male, 22 female, 16-47 y | Handball | 331,329 | Self-report | Shoulder |
| Mohammadi, 2007[41] | Male only[b], 22-26 y | Soccer | 20,20 | Staff report | Ankle |
| Cotellessa 2023[12] | Male only, 16.9 ± 0.7 y | Soccer | 21,21 | Staff report | Groin |
| Shitara, 2022[52] | Male only, 15-17 y | Baseball | 51,62 | Self-report | Shoulder |
| Åkerlund, 2020[3] | Sex unknown, 12-17 y | Floorball | 301,170 | Self-report | Overall |
| Raya-González, 2021[49] | Male only, <19 y | Soccer | 22,22 | Staff report | Muscle |
| Harøy, 2019[21] | Male only, 22.84 ± 4.38 y[c] | Soccer | 247,242 | Self-report | Groin |
| Achenbach, 2018[2] | Sex unknown, <16-18 y | Handball | 168,111 | Staff report | Overall |
| Achenbach, 2022[1] | 320 male, 259 female, senior/U-19 | Handball | 284,295 | Self-report | Shoulder |
| van der Horst, 2015[56] | Male only, 24.5 ± 3.8 y | Soccer | 292,287 | Self-report | Hamstring |
| Hölmich, 2010[25] | Male only, Age unknown | Soccer | 477,430 | Self-report | Groin |
| Hasebe, 2020[23] | Male only, 15-18 y | Soccer | 156,103 | Staff report | Overall |
| Stojanović, 2023[54] | 6 male, 2 female, 21.6 ± 2.6 y | Basketball | 57,55 | Staff report | Low limb |
| Nuhu, 2021[42] | Male only, 20 ± 1 y | Soccer | 312,318 | Staff report | Low limb |
| Soligard, 2008[53] | Female only, 13-17 y | Soccer | 1055,837 | Self-report | Low limb |
| Petersen, 2011[47] | Male only, 19-28 y | Soccer | 461,481 | Self-report | Hamstring |

[a]Indicates number of teams of that sex, e.g.: 23 male means 23 male teams.
[b]Exact number given in sample size.
[c]Combined means and SD.

strength training on these injuries, and concluded that the significance of strength training in the mitigation of sports injuries suggests the need to consider arranging more strength training exercises into IPPs.

Above all, we need to recognize the relationship between different contact sports and their common injuries. The occurrence of groin and hamstring injuries is related to the athletes' running and changing direction on the field. For some sports we included, such as soccer and floorball, the weaker the adductor or hamstring muscles, the greater the risk of injury. However, more strength in a particular site does not mean less risk of injury in that site. Even for the same person, an injury in a specific location of the body has different reasons, because the motor function of the different locations of the body is specific to that location. Some sports, such as basketball and handball, require athletes to complete a jumping action. It is posited that a comprehensive jump comprises a sequence of actions, including braking, jumping, landing, and buffering. During this process, the knee and ankle joints are particularly susceptible to injury.[24] In addition, we believe that in sports that require a jump action, the athlete's movement patterns, body balance, and joint flexibility or stability have a more significant effect on the incidence of ankle and knee injuries than does strength. Consequently, it is imperative to examine whether strength training can mitigate the risk of sports injuries by targeting specific body regions. Additionally, it is essential to develop strength IPPs with tailored components for various injury-prone areas.

On the basis of the above, along with previous research,[33] we conclude that acute injury results from a series of dynamic movements during exercise, including changing direction and sudden stops and starts. Furthermore, certain contact sports necessitate repetitive movements such as jumping and throwing, which may lead to an increased risk of overuse injuries.[48] Consequently, it is evident that injuries at different anatomic locations

TABLE 3
Characteristics of Interventions

| First Author, y | Intervention | Compliance, % | Exposure Time | Frequency | Sets, Repetitions (reps) |
|---|---|---|---|---|---|
| Andersson, 2017[4] | Shoulder strength | /[a] | 7 mo | 3/wk | / |
| Mohammadi, 2007[41] | Ankle strength | / | 32 wk | 7/wk | 10 sets<br>20 reps |
| Cotellessa, 2023[12] | Adductor strength | 81 | 24 wk | 2/wk | 2.33 sets<br>8.83 reps |
| Shitara, 2022[52] | Shoulder strength | / | 150 d | 7/wk | 20 reps<br>10 s |
| Åkerlund, 2020[3] | Knee strength | 83 | 26 wk | 7/wk | Unclear |
| Raya-González, 2021[49] | Core and hamstring strength | 92.3 | 10 wk | 2/wk | 2-3 sets<br>5-12 reps |
| Harøy, 2019[21] | Adductor strength | 73 | 6-8 + 28 wk | 2-3/wk + maintain 1/wk | 2 sets<br>8.2-10.8 reps |
| Achenbach, 2018[2] | Shoulder multicomponent | / | 10-12 wk + 1 season | 2-3/wk + maintain 1/wk | / |
| Achenbach, 2022[1] | Multicomponent | 78 | 1 season | 2-3/wk | 2-3 sets<br>8-10 reps |
| van der Horst, 2015[56] | Hamstring strength | 91 | 13 wk | 2/wk | 2-3 sets<br>6.5-7.5 reps |
| Hölmich, 2010[25] | Multicomponent | / | 33 wk | 2-4/wk | / |
| Hasebe, 2020[23] | Hamstring strength | 88 | 27 wk | 2/wk | 5-10 sets<br>2-3 reps |
| Stojanović, 2023[54] | Multicomponent | 75 | 6 mo | 3-4/wk | / |
| Nuhu, 2021[42] | Multicomponent | 77 | 7 mo | 2.8-4.9/wk | / |
| Soligard, 2008[53] | Multicomponent | 77 | 1 season | 2-5/wk | 3 sets<br>7.33-10 reps |
| Petersen, 2011[47] | Hamstring strength | 91 | 12 mo | 2.7/wk + maintain 1/wk | 2-3 sets<br>8.1-10.2 reps |

[a]Virgule indicates unknown or unreported.

have distinct influencing factors (Figure 10), necessitating the development of targeted IPPs.

## Single-Component Strength Training Reduces Site-Specific Muscle Injuries

The findings of the meta-analysis conducted in this study indicate that the efficacy of single-component strength training in injury prevention is site-specific, demonstrating more favorable outcomes in the prevention of groin and hamstring injuries. However, when designing single-component strength training to effectively reduce sports injuries, we still need to mention a few matters.

First, although single-component strength training only includes the strength of this training component, it does not mean that we only carry out 1 strength training exercise. When we design a set of single-component strength training lessons, we can consider combining various forms of strength such as concentric, eccentric, isometric, and so forth, which may have better results; and the sets and repetitions in strength training can be determined according to the actual situation of the athletes and obey the principle of gradual progress. Second, besides strength training itself, we also need to pay attention to the influence of external factors on the strength training effect, such as training schedules, temperature and so on. Above, we mentioned that site-specific single-component strength training can play a role in injury prevention, but it is important to understand why single-component strength training works. We analyzed the structure of groin and hamstring injuries, which can be attributed to injury of the muscles in these 2 areas more precisely. From the anatomic structure, we can see that these 2 muscles belong to the superficial and large muscles that produceactive force, and most of the injuries are strains, tears, inflammation, and other conditions. Both injuries are very common in sports.

Groin injuries account for 8% to 18% of all injuries in football, and the injured typically miss about 15 days of training after a groin injury.[58] We believe that athletes' groin injuries during competitions are very detrimental to the match outcome, especially for important athletes, whose athletic performance has an important effect on the performance of the entire team. Therefore, we conclude that single-component strength training for the strength of large muscle groups can reduce sports injuries. In addition,

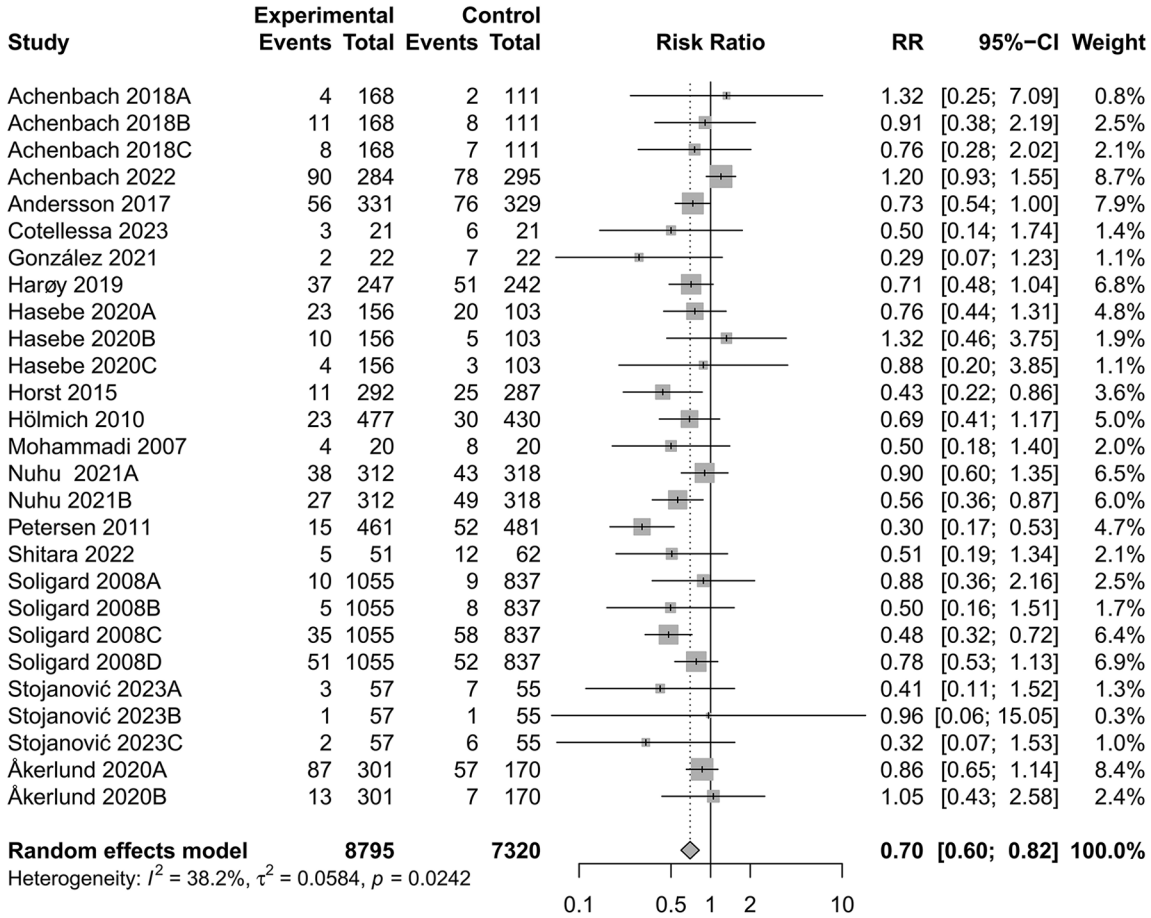


**Figure 2.** Effect of strength training on reducing sports injury rate. RR, relative risk.

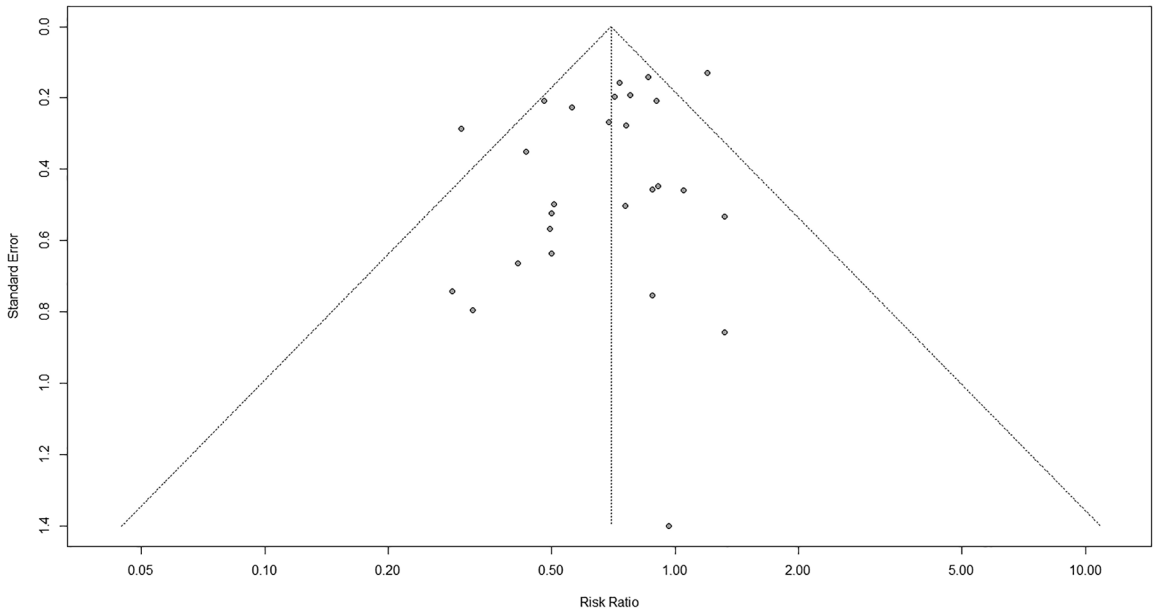


**Figure 3.** Funnel plot.

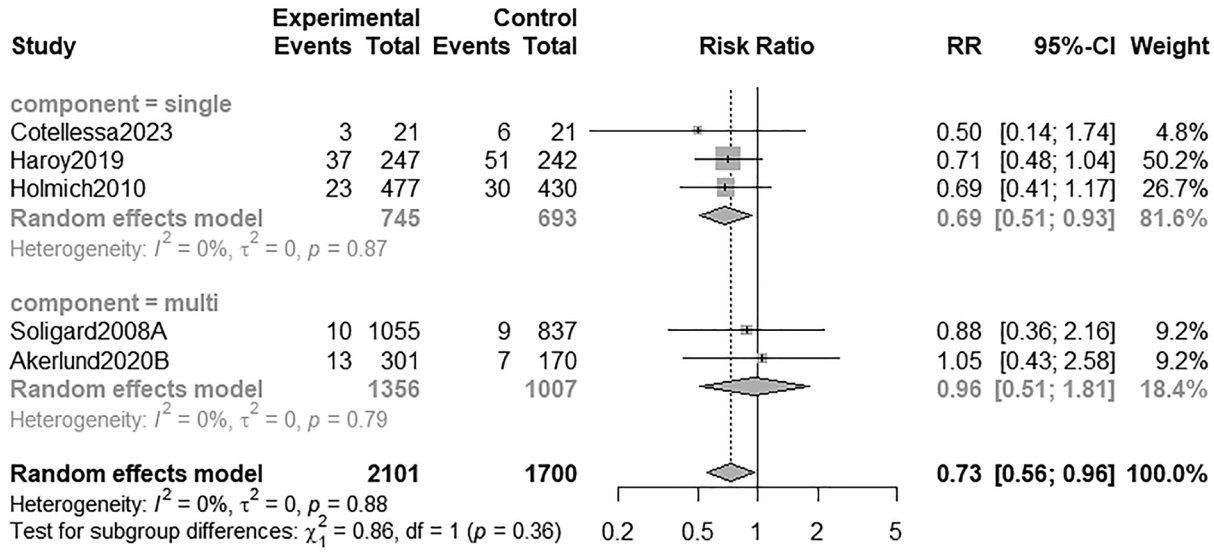


**Figure 4.** Effect of single-component strength training on reducing groin injuries. RR, relative risk.

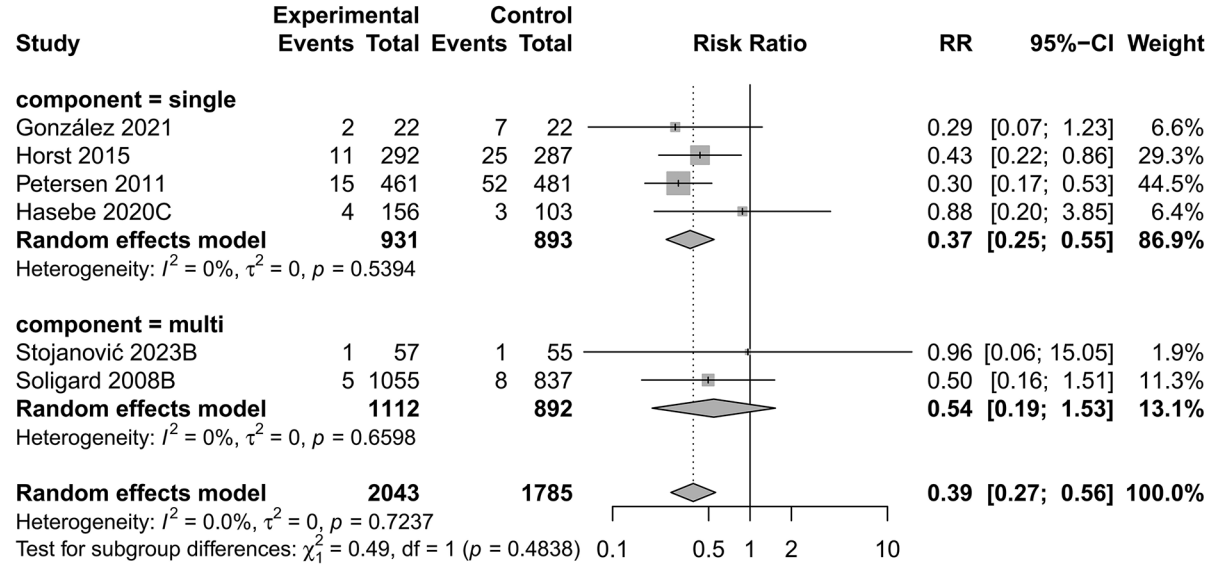


**Figure 5.** Effect of single-strength training on reducing hamstring injuries. RR, relative risk.

there is a study[33] suggesting that implementing progressive hip joint strength training in young football players can be an effective preventive measure against such injuries. According to this current study, this effect is likely due to the improved strength of the adductor muscle, but the role of strength training of other muscles around the hip remains to be investigated.

Moreover, the current study found that single-component strength training reduced the risk of hamstring injury, although this is not a new idea, as there have been numerous studies that support it. First, during athletic activities, the biomechanical demands placed on the hamstring muscles are unique; they are required to be activated when performing both hip extension and knee flexion.[13] This dual role makes them particularly susceptible to strains, especially during high-velocity actions such as sprinting and jumping.[17,19] Second, we believe that when there is an imbalance in muscle strength, the tension on the muscles is more likely to cause hamstring injury. For instance, the research findings indicate that thigh muscle

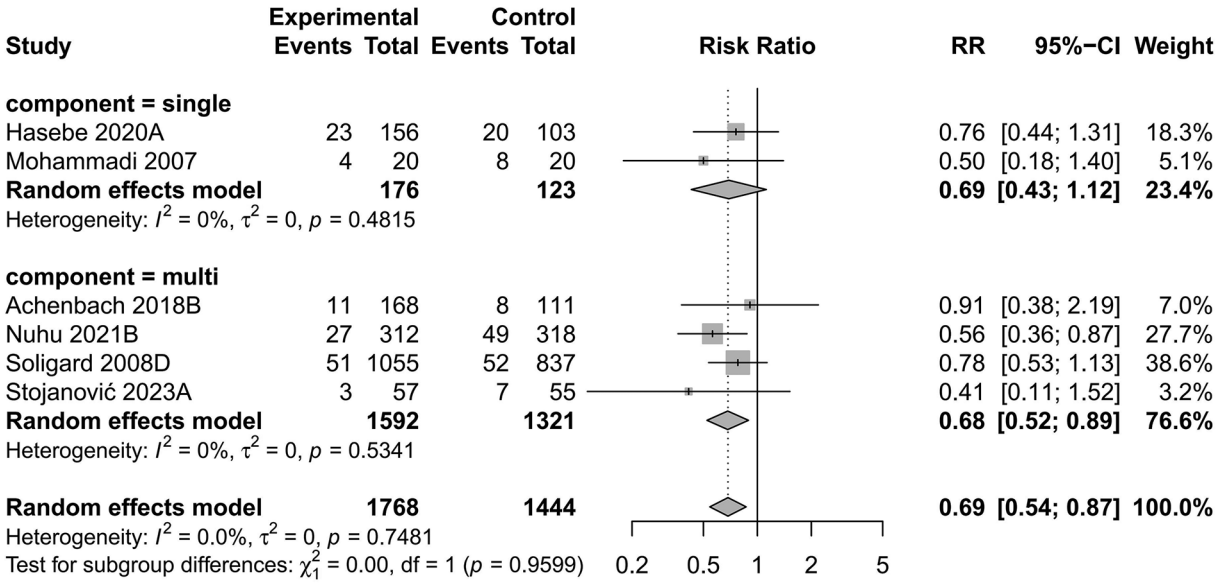


**Figure 6.** Effect of multicomponent strength training on reducing ankle injuries. RR, relative risk.

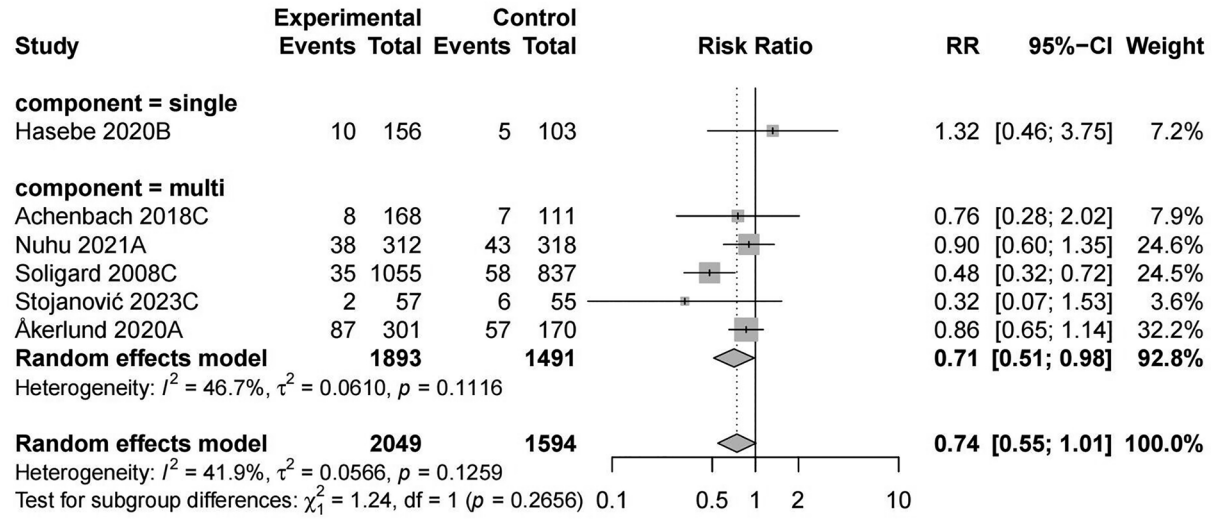


**Figure 7.** Effect of multicomponent strength training on reducing knee injuries. RR, relative risk.

strength imbalances exceeding 15% and 20% bilaterally elevate the risk of hamstring injuries by 2.4 times and 3.4 times, respectively.[9] Remarkably, strength training has been demonstrated to enhance hamstring strength and promote optimal tendon length, leading to a notable decrease in peak hamstring tension during sprinting and consequently lowering the likelihood of hamstring injuries.[57] Therefore, enhancing thigh muscle strength imbalances via strength training has been shown to be an effective strategy for mitigating the risk of hamstring injuries.

## Multicomponent Strength Training Reduces Site-Specific Joint Injuries

Our study's results suggest that multicomponent strength training reduces injury rates of both the knee and the ankle in contact sport athletes. As both injuries are directly related to the joint, their structures are more complex. Therefore, compared with the 2 muscle injuries mentioned above (groin and hamstring), the factors affecting the knee joint and ankle joint are more numerous and

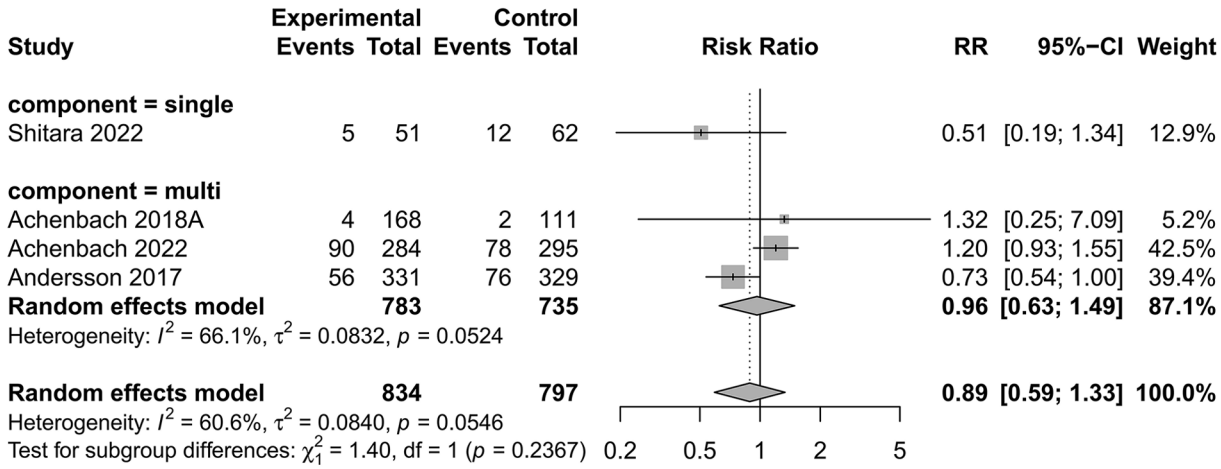


**Figure 8.** Effect of strength training on reducing shoulder injuries. RR, relative risk.

more complex, and multicomponent strength training can be an important means to prevent these injuries.

Research suggests that multicomponent strength training is a beneficial strategy for reducing ankle joint injuries through enhancements in neuromuscular control, proprioception, muscle strength balance, and core stability.[26] Theoretically, different exercises can reduce the risk of injury from different causes. And for the strength training component, studies have suggested that enhancing the musculature surrounding the knee joint can provide a beneficial safeguard against neuromuscular impairments associated with anterior cruciate ligament (ACL) injuries.[43] However, certain investigations propose that strength training may provide a lesser effect in terms of enhancement of landing biomechanics compared with alternative training modalities.[45] Further, for ankle injuries, previous studies on ankle joint injuries also have indicated that a 6-week ankle joint strength training program did not yield statistically significant improvements in ankle joint strength.[31] Therefore, we conclude that single-component strength training is not very effective in preventing complex joint injuries, possibly because complex joint injuries are also affected by other factors such as movement patterns, and some joint strength training effects are not significant. For knee and ankle injuries, balance and proprioception training that are included in multistrength training improved proprioceptive sensation, neuromuscular control, joint stability, and so forth.[27,37] In addition, multistrength training includes plyometric training that can improve landing mechanics, reduce maximal landing forces, and reduce ACL injuries.[32,45] The effectiveness of these training modalities is further supported by evidence indicating that they can improve landing mechanics, which is crucial for reducing knee and ankle injuries.[10]

In conclusion, multicomponent IPPs demonstrate advantages in reducing the incidence of ankle and knee injuries among athletes. The effective IPPs should be comprehensive, incorporating multiple components, necessitating sustained and consistent adherence to protocol, and attaining a certain level of proficiency to mitigate the risk of injury. As such, these training instructions should be emphasized in athletic training programs to promote safer participation in sports.

## Inapparent Effect of Strength Training on Reducing Shoulder Injuries

Our meta-analysis showed that strength training did not reduce the incidence of shoulder injuries, other injuries. Theoretically, the shoulder is particularly vulnerable during overhead sports, and merely enhancing muscle strength may not adequately mitigate injury risk without addressing other factors such as biomechanics and training load.[11] The findings from Krajnik et al[34] further support this notion, indicating that a focus on biomechanics and conditioning is crucial for preventing shoulder injuries in baseball athletes. Moreover, Shitara et al[52] conducted a study comparing shoulder stretching and strength training for injury prevention among high school pitchers, finding that the strength training did not significantly lower injury incidence compared with stretching.

The systematic review by Asker et al[5] noted that shoulder injuries in overhead sports such as volleyball are multifactorial, involving not only strength but also flexibility and neuromuscular control.[5] Moreover, Forthomme et al[16] found no significant association between shoulder strength assessments and injury occurrence among elite handball players, suggesting that strength alone may not be a reliable predictor of injury risk.

In conclusion, the multifaceted nature of shoulder injuries necessitates a more comprehensive approach that includes not only strength training but also flexibility, biomechanics, and neuromuscular training to effectively mitigate injury risks.

## Limitations and Future Suggestions

This research studied the injury prevention effect of different components of strength training in contact sports,

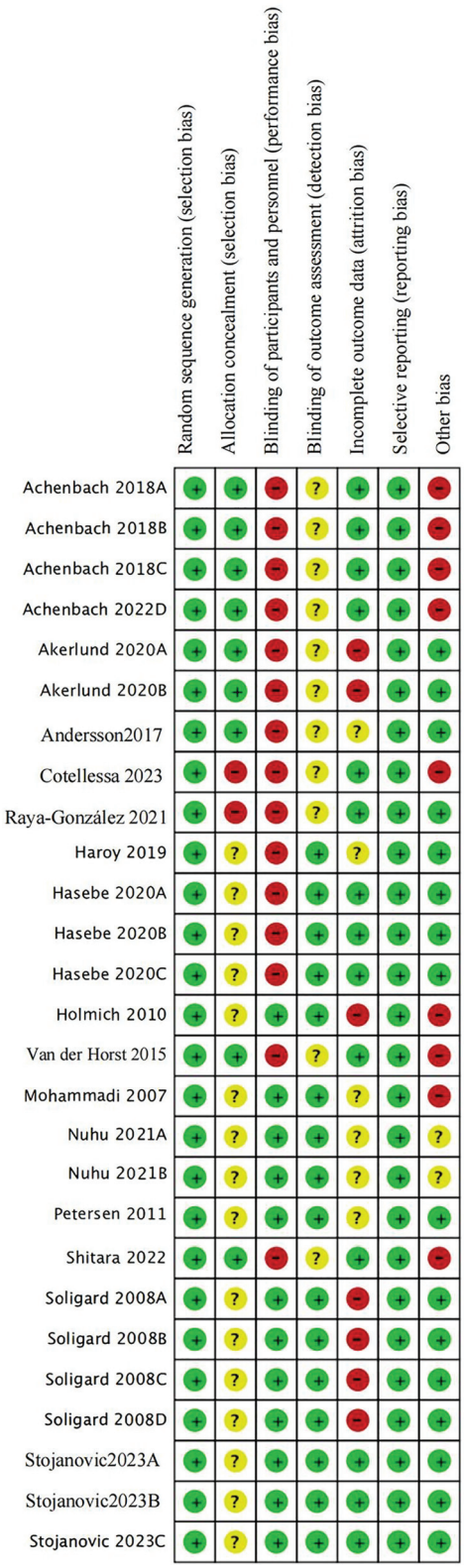


**Figure 9.** Literature quality assessment chart.

utilizing a previously established definition of contact sports. But the discussion of numerous sports that satisfied the definition of "contact sport" are still insufficient due to the limited availability of literature. Table 1 introduces the types of trauma primarily focusing on noncontact injuries that occur without the involvement of external objects. However, outside of the 5 injuries included in this study, there are many other prevalent injuries in contact sports, such as lower back trauma and elbow joint injuries.

This study categorizes strength training into 2 types: multicomponent and single-component. Although all included studies identified their interventions as strength training, the definitions of strength training varied across the literature. During the study selection process, literature was screened based on reported outcome indicators. Consequently, many studies relevant to the research objective were excluded because they did not report injury rates as an outcome. Such inclusion criteria may result in the omission of valuable literature. In addition, the lack of comprehensive data regarding the appropriate amount of strength training has been recognized as a critical factor influencing the effectiveness of strength training programs in preventing injury. To theoretically investigate this issue, metaregression analysis was employed to examine the effect of varying strength training applications on the incidence of sports injury. In practice, strength training typically employs progressive loading, and its training frequency, number of sets, and number of repetitions are determined by different training periods and athlete competence levels. In addition, some studies have not reported the load and volume of strength training, and the forms of strength training used may be different.

In light of the limitations identified in our research, we propose several directions for future studies. First, it is imperative to gather additional evidence, encompassing investigations into injury prevention across diverse locations, sports, and outcome measures. Second, there is a need to explore varying applications of strength training, specifically training frequency, number of sets, and number of repetitions. Furthermore, it is crucial to consider the distinct forms of strength training—concentric, eccentric, and isometric—and to analyze their preventive effects on injuries in various locations of the body. This includes examining the underlying causes and potential mechanisms contributing to observed differences.

## CONCLUSION

There are some differences between multicomponent strength training and single-component strength training in reducing sports injury. These differences may be caused by the anatomic structure and function of the injured sites, which suggest that the effect of single-component strength training is more effective for hamstring and groin injuries, and multicomponent strength training is more effective in preventing ankle and knee injuries. In addition, for ankle

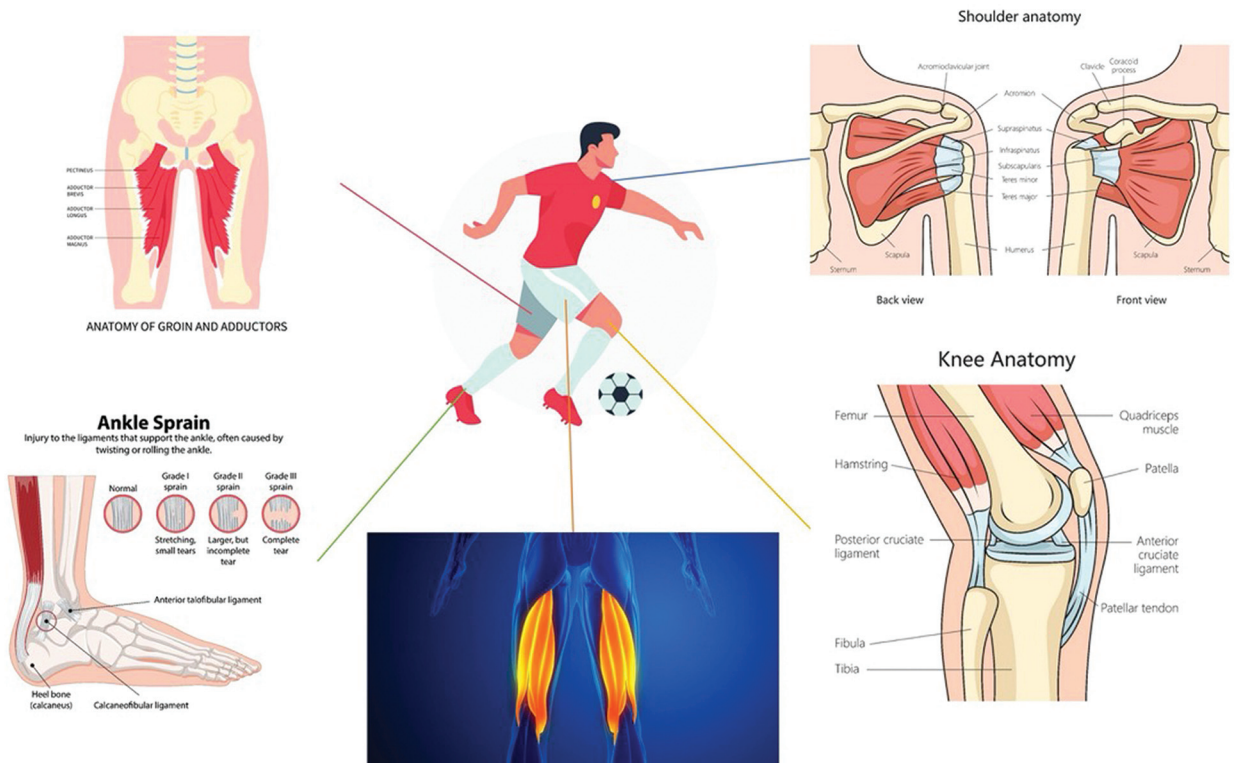


**Figure 10.** Anatomic diagram of each injury site.

and knee injuries, strength training should be combined with balance, coordination, and other exercises related to improved movement patterns, which cannot be neglected in arranging IPPs.

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